

EXERCISE SHEET ① (MATH 124)

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CONICS PARAMETRIC EQUATIONS AND POLAR COORDINATES

Ex 1 A curve C is defined by the parametric equations $x=t^2$
 $y=t^3-3t$

- Show that C has two tangents at point $(3,0)$ and find their equations
- Find the points on C where tangent is horizontal and vertical
- Determine where the curve is concave upward or downward
- Sketch the curve

① $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2-3}{2t}$ let's find how can we obtain

the point $(3,0)$? $t^2=3$ $t^3-3t=0$
 $t=0$ $t=\pm\sqrt{3}$
 $t(t^2-3)=0$ $t=0$ $t=\pm\sqrt{3}$
This gives $(0,0)$ point

So, we have two parameters to get $(3,0)$

then $\frac{dy}{dx} \Big|_{t=\sqrt{3}} = \sqrt{3}$

$\frac{dy}{dx} \Big|_{t=-\sqrt{3}} = -\sqrt{3}$

Equation 1

$y-0 = \sqrt{3}(x-3)$

Equation 2

$y-0 = -\sqrt{3}(x-3)$

② Horizontal, $\frac{dy}{dt} = 0$ $3t^2-3=0$ $t=\pm 1$ then (Note $\frac{dx}{dt} \neq 0$ there)

we have $t=1$ $x=1$ $y=-2$ $(1,-2)$
 $t=-1$ $x=1$ $y=2$ $(1,2)$

horizontal tangents at points $(1,-2)$ $(1,2)$

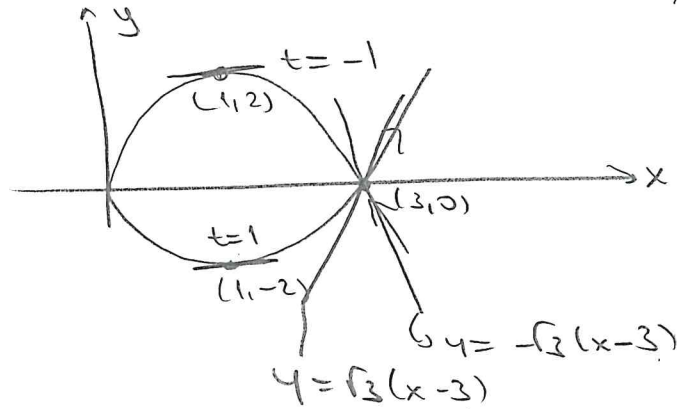
Vertical $\frac{dx}{dt} = 0$ $t^2=0$ (Note $\frac{dy}{dt} \neq 0$) $t=0$

we have vertical tangent at $(0,0)$

③ Concavity $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) / \frac{dx}{dt}$ $\frac{dy}{dx} = \frac{3}{2} \left(t - \frac{1}{t} \right)$

$\frac{d^2y}{dx^2} = \frac{3/2 (1 + 1/t^2)}{2t} = \frac{3(t^2+1)}{4t^3}$ $t > 0$ upward
 $t < 0$ downward

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$x=t^2 \Rightarrow$ symmetry w.r.to t.

Ex 2 Find $\frac{dy}{dx}$ for the following functions

a) $x = t - t^3$ $y = 2 - 5t$

b) $x = t e^t$ $y = t + e^t$

a) $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-5}{1-3t^2}$ b) $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1+e^t}{e^t + t e^t}$

Ex 3 Find an equation of the tangent to the curve $x = \cos \theta + \sin 2\theta$
 $y = \sin \theta + \cos 2\theta$ at $\theta = 0$.

Sol $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\cos \theta - 2 \sin 2\theta}{-\sin \theta + 2 \cos 2\theta}$

$\left. \frac{dy}{dx} \right|_{\theta=0} = \frac{1}{2}$ (slope)

Equation $\Rightarrow y(0) = 1$ $y - 1 = \frac{1}{2}(x - 1)$
 $x(0) = 1$

Ex 4 For the following curves obtain $\frac{d^2y}{dx^2}$.

a) $x = t - e^t$ $y = t + e^{-t}$ b) $x = t + \ln t$ $y = t - \ln t$

a) $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1 - e^{-t}}{1 - e^t} = \frac{e^t - 1}{1 - e^t} = -e^{-t}$

$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} = \frac{e^{-t}}{1 - e^t}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1 - 1/t}{1 + 1/t} = \frac{t-1}{t+1}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) / \frac{dx}{dt} = \frac{(1+t) - (t-1)}{(1+t)^2} = \frac{2}{(1+t)^2} = \frac{2t}{(1+t)^3}$$

Ex 5 Find the points of the curve $x = 2\cos\theta$ $y = \sin 2\theta$.

where the tangent is horizontal or vertical $0 \leq \theta \leq 2\pi$

Vertical

$$\frac{dx}{d\theta} = -2\sin\theta \quad \frac{dx}{d\theta} = 0 \quad -2\sin\theta = 0 \quad \theta = n\pi \text{ so}$$

we have vertical tangents at $\theta = n\pi \Rightarrow (2, 0), (-2, 0)$

$$\text{Horizontal } \frac{dy}{d\theta} = 2\cos 2\theta \quad \frac{dy}{d\theta} = 0 \quad \cos 2\theta = 0 \quad \theta = (2n+1)\pi/4$$

So we have horizontal tangent lines at point $(0, 1), (0, -1)$

Ex 6 Find an equation of the tangent to the curve.

$$x = e^{\sqrt{t}} \quad y = t - \ln t^2 \quad \text{at point } t = 1$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1 - \frac{2}{t}}{\frac{1}{2\sqrt{t}} e^{\sqrt{t}}} = \frac{1 - \frac{2}{t}}{\frac{e^{\sqrt{t}}}{2\sqrt{t}}} \quad \frac{dy}{dx} \Big|_{t=1} = \frac{-1}{e}$$

$$t = 1 \quad x = e \quad y = 1 - \ln 1 = 1 \quad (e, 1)$$

$$\text{Equation } x - e = -\frac{2}{e} (y - 1)$$

Ex 7 At what points on the curve $x = t^3 + 4t$ $y = 6t^2$ is the tangent parallel to the line with equation $x = -7t$ $y = 12t - 5$

$$\text{the slope of the curve } x = -7t \quad y = 12t - 5 \quad \frac{dy}{dx} = \frac{12}{-7}$$

$$\text{Our aim to find } \frac{dy}{dx} = -\frac{12}{7} \text{ for } x = t^3 + 4t \quad y = 6t^2$$

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$$x = t^3 + 4t \quad y = 6t^2 \quad \frac{dy}{dx} = \frac{12t}{3t^2 + 4} \quad \frac{dy}{dx} \text{ will be equal to } -\frac{12}{7}$$

$$\frac{dy}{dx} = \frac{12t}{3t^2 + 4} = -\frac{12}{7} \quad \text{So, } \cancel{12} \cdot t = -\cancel{12} (3t^2 + 4)$$

$$3t^2 + 7t + 4 = 0 \quad (3t+4)(t+1) = 0$$

$$\begin{array}{ccc} 3t & + & 4 \\ t & + & 1 \end{array} \quad \begin{array}{cc} \downarrow & \downarrow \\ t = -4/3 & t = -1 \end{array}$$

So for $t = -1$ we have $(x, y) = (-5, 6)$
 " $t = -4/3$ " " $(-\frac{208}{27}, \frac{132}{3})$

Ex 7 Set up, DO NOT EVALUATE, an integral that represents the length of the given curve

a) $x = t - t^2 \quad y = \frac{4}{3}t^{3/2} \quad 1 \leq t \leq 2$

b) $x = \ln t \quad y = \sqrt{t+1} \quad 1 \leq t \leq 5$

a) $\frac{dx}{dt} = 1 - 2t \quad \frac{dy}{dt} = 2 \cdot t^{1/2}$

$$L = \int_1^2 \sqrt{(1-2t)^2 + 4t} \, dt = \int_1^2 \sqrt{1+4t^2} \, dt$$

b) $\frac{dx}{dt} = \frac{1}{t} \quad \frac{dy}{dt} = \frac{1}{2\sqrt{t+1}}$

$$L = \int_1^5 \sqrt{\frac{1}{t^2} + \frac{1}{4(t+1)}} \, dt$$

Ex 8 Find the length of the curve $x = a(\cos \theta + \theta \sin \theta)$
 $y = a(\sin \theta - \theta \cos \theta) \quad 0 \leq \theta \leq \pi \quad a > 0$

$$\frac{dx}{d\theta} = a(-\sin \theta + \theta \cos \theta + \sin \theta) = a\theta \cos \theta$$

$$\frac{dy}{d\theta} = a(\cos \theta - \cancel{\cos \theta} + \theta \sin \theta) = a\theta \sin \theta$$

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = a^2 \theta^2 (\cos^2 \theta + \sin^2 \theta)$$

$$L = \int_0^\pi \sqrt{a^2 \theta^2} \, d\theta = \int_0^\pi a\theta \, d\theta = a \frac{\theta^2}{2} \Big|_0^\pi = a \frac{\pi^2}{2}$$

Ex 9 Find the area of the following surfaces

a) $x = t^3$ $y = t^2$ $0 \leq t \leq 1$ Rotate x axis

b) $x = e^t - t$ $y = 4e^{t/2}$ $0 \leq t \leq 1$ Rotate around y axis

a) $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (3t^2)^2 + (2t)^2 = 9t^4 + 4t^2$

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \sqrt{9t^4 + 4t^2} dt$$

Rotate x $\int_0^1 2\pi y ds = \int_0^1 2\pi \cdot \underbrace{t^2}_{t^2 \cdot t \sqrt{9t^2+4}} \sqrt{9t^4+4t^2} dt$

$$9t^2 + 4 = u \quad t^2 = \frac{u-4}{9}$$

$$= 2\pi \int_4^{13} \left(\frac{u-4}{9}\right) \sqrt{u} \left(\frac{1}{6}\right) du$$

$$du = 18t dt \quad t dt = \frac{du}{18}$$

$$= \frac{2\pi}{9 \cdot 18} \int_4^{13} (u^{3/2} - 4u^{1/2}) du = \frac{2\pi}{81}$$

$$t=0 \Rightarrow u=4$$

$$t=1 \Rightarrow u=13$$

$$= \frac{\pi}{81} \left[\frac{2}{5} u^{5/2} - \frac{8}{3} u^{3/2} \right]_4^{13} = \frac{2\pi}{1215} (247\sqrt{13} + 64)$$

b) $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

$$ds = \sqrt{(e^t - 1)^2 + (2 \cdot e^{t/2})^2}$$

$$= \sqrt{e^{2t} - 2e^t + 1 + 4e^t}$$

Rotate y $\int_0^1 2\pi x ds =$

$$= \sqrt{e^{2t} + 2e^t + 1} = e^t + 1$$

$$\int_0^1 2\pi (e^t - t) (e^t + 1) dt = 2\pi \int_0^1 (e^{2t} + e^t - te^t - t) dt$$

$$= 2\pi \left(\frac{e^{2t}}{2} + e^t - (t-1)e^t - \frac{t^2}{2} \right) \Big|_0^1 = \pi(e^2 + 2e - 6)$$

$$\int te^t dt = e^t \cdot t - \int e^t dt = e^t(t-1)$$

$$t=u \quad dt=du$$

$$e^t dt = dv \quad e^t = v$$

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Ex 10 Find the equations of following curves

(a) Ellipse with foci at $(0, \pm 1)$ and semi major axis 2.

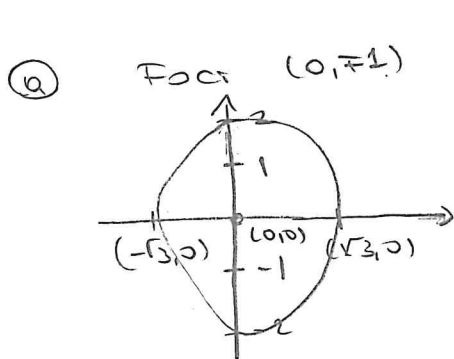
(b) Ellipse foci $(0, \pm 1)$ and $(4, \pm 1)$ and semi major axis 5

(c) Parabola with focus $(2, 3)$ and vertex at $(2, 4)$

(d) Obtain the equation of parabola with focus at $(0, -1)$ and vertex $(\frac{1}{2}, 1)$ using distance between points, and prove that it passes through the origin.

(e) hyperbola with foci at $(0, \pm 2)$ and with vertices $(\pm 1, 0)$

(f) hyperbola with foci $(\pm 5, 1)$ and asymptotes $y = \pm (y-1)$



Vertices $(0, \pm 2)$
 Covertices $(\pm \sqrt{3}, 0)$
 Semi major 2
 Semi minor $\sqrt{3}$

$$\frac{x^2}{(\sqrt{3})^2} + \frac{y^2}{2^2} = 1$$

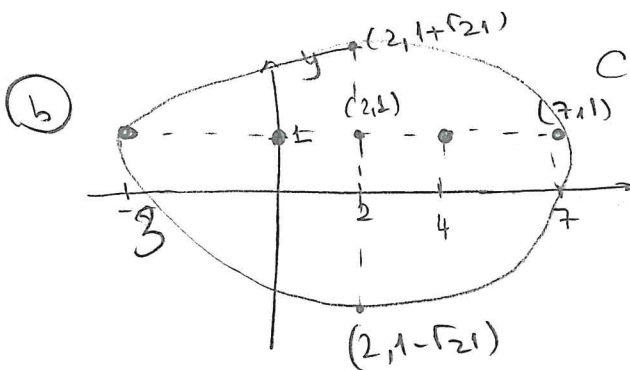
$$\frac{x^2}{3} + \frac{y^2}{4} = 1$$

$$c = \pm 1 \quad a = \pm 2$$

$$c^2 = a^2 - b^2$$

$$1 = 4 - b^2$$

$$b = \pm \sqrt{3}$$



Center point halfway between foci.

S-o center $(2, 1)$ $c = 2$
 Semi major 5 $a = 5$

$$c^2 = a^2 - b^2 \quad 4 = 25 - b^2$$

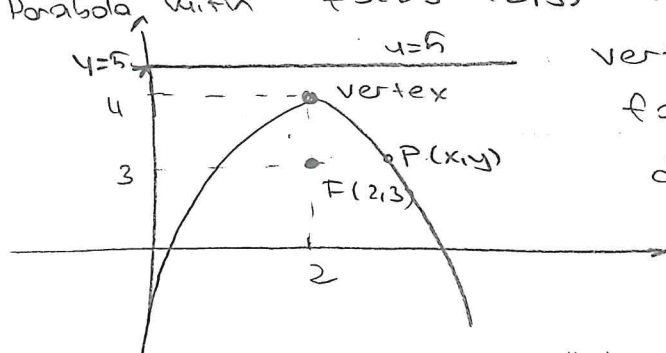
$$b^2 = 21 \quad b = \pm \sqrt{21}$$

Vertices $(7, 1)$ $(-3, 1)$

Covertices $(2, 1+\sqrt{21})$ $(2, 1-\sqrt{21})$

$$\frac{(x-2)^2}{25} + \frac{(y-1)^2}{21} = 1$$

② Parabola with focus (2,3) and vertex (2,4)



vertex $(h, k) = (2, 4)$ $k = 4$
 focus $(h, k+p) = (2, 3)$ $p = -1$
 directrix $y = 4 - (-1) = 5$.

Equation in general will be in the form $(x-2)^2 = -4(y-4)$

Obtain it. $|PF| = \sqrt{(x-2)^2 + (y-3)^2}$

Directrix distance $|5-y|$

$$\sqrt{(x-2)^2 + (y-3)^2} = (5-y)$$

$$(x-2)^2 + (y-3)^2 = 25 - 10y + y^2$$

$$(x-2)^2 = 25 - 10y + y^2 - y^2 + 6y - 9$$

$$(x-2)^2 = 16 - 4y = -4(y-4)$$

③ Vertex $(1/2, -1) = (h, k)$ $h = 1/2$ $k = -1$

Focus $(0, -1) = (h+p, k)$ $p = -1/2$

Directrix $h-p = 1 = x$

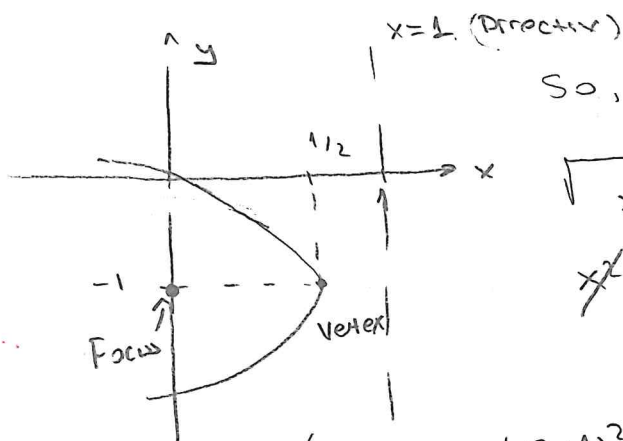
Then we get the equation $(y+1)^2 = 4(-1/2)(x-1/2)$
 $= -2(x-1/2)$

④ Obtain using distance formula

Let $P(x, y)$ be any point on the parabola, then.

$$|PF| = \sqrt{x^2 + (y+1)^2}$$

We have directrix $x=1$ distance between P and directrix $|x-1|$



So, we have

$$\sqrt{x^2 + (y+1)^2} = |x-1|$$

$$x^2 + (y+1)^2 = x^2 - 2x + 1$$

$$(y+1)^2 = -2(x-1/2)$$

It satisfies $(0,0)$ $(0+1)^2 = -2(0-1/2) \Rightarrow 1=1$

So it satisfies the condition (7)

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(e) (I) Foci $(0, \pm 2)$ $\Rightarrow c = 2$ $c^2 = a^2 + b^2$
 Vertices $(0, 1)$ $(0, -1) \Rightarrow a = 1$ $4 = 1 + b^2$ $b^2 = 3$
 So, we will have $\frac{y^2}{1} - \frac{x^2}{3} = 1$

(II) Assume $P(x, y)$ on the hyperbola $F_1 = (0, 2)$ $F_2 = (0, -2)$
 $|PF_1| = \sqrt{x^2 + (y-2)^2}$ $|PF_2| = \sqrt{x^2 + (y+2)^2}$

$$\sqrt{x^2 + (y-2)^2} - \sqrt{x^2 + (y+2)^2} = \pm 2 \quad (\pm 2a)$$

$$\cancel{x^2} + (y-2)^2 = 4 \pm 4\sqrt{x^2 + (y+2)^2} + \cancel{x^2} + (y+2)^2$$

$$\cancel{y^2} - 4y + \cancel{4} = 4 \pm 4\sqrt{x^2 + (y+2)^2} + \cancel{y^2} + 4y + \cancel{4}$$

$$-4y + 4 \pm 4\sqrt{x^2 + (y+2)^2} = 0$$

$$-y + 1 \pm \sqrt{x^2 + (y+2)^2} = 0$$

$$x^2 + (y+2)^2 = 4y^2 + 4y + 1$$

$$x^2 + y^2 + 4y + 4 = 4y^2 + 4y + 1$$

$$4y^2 - y^2 - x^2 = 3$$

$$3y^2 - x^2 = 3$$

$$y^2 - \frac{x^2}{3} = 1$$

(f) Foci $(\pm 5, 1)$ asymptotes $x = \pm (y-1)$

For $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ $y-k = \pm \frac{b}{a}(x-h)$ Center (h, k)
 $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$ $y-k = \pm \frac{a}{b}(x-h)$

we have asymptote $x = \pm (y-1)$ $(h, k) = (0, 1)$
 $a/b = 1$ $a = b$ $c^2 = a^2 + b^2$

Center $(0, 1)$

Foci $(0, +5, 1)$

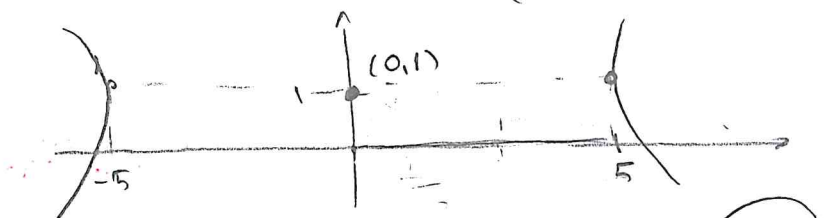
$(0, -5, 1)$

$c = 5$ $25 = 2a^2$ $a^2 = \frac{25}{2}$

$$\frac{x^2}{\frac{25}{2}} - \frac{(y-1)^2}{\frac{25}{2}} = 1$$

$$x^2 - (y-1)^2 = \frac{25}{2}$$

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(8)

Ex 11 Find the equation of the ellipse with foci $(2, -2), (4, -2)$ and vertices $(1, -2), (5, -2)$.

(a) By using the general representation of ellipses

(b) By using the distance formula between points

(a) The major axis that joins the vertices is at x line.

$(1, -2), (5, -2)$. Length 4 so $a = 2$. The distances

$$5 - 1 = 4$$

between foci $(4, -2), (2, -2)$ is 2 so $c = 1$

$$4 - 2 = 2$$

Thus, $a^2 = b^2 + c^2$ $4 = b^2 + 1$ $b = \sqrt{3}$

The center of the ellipse is the middle of foci, then $(3, -2)$, so we have $(h, k) = (3, -2)$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \Rightarrow \frac{(x-3)^2}{4} + \frac{(y+2)^2}{3} = 1$$

(b) Let P be any point at ellipse $P(x, y)$ then

$$|PF_1| + |PF_2| = 2a, \quad a = 2$$

$$\sqrt{(x-2)^2 + (y+2)^2} + \sqrt{(x-4)^2 + (y+2)^2} = 4$$

$$(x-2)^2 + (y+2)^2 = 16 - 8\sqrt{(x-4)^2 + (y+2)^2} + (x-4)^2 + (y+2)^2$$

$$x^2 - 4x + 4 = 16 - 8\sqrt{(x-4)^2 + (y+2)^2} + x^2 - 8x + 16$$

$$28 - 4x = -8\sqrt{(x-4)^2 + (y+2)^2}$$

$$7 - x = -2\sqrt{(x-4)^2 + (y+2)^2}$$

$$(7-x)^2 = 4(x-4)^2 + 4(y+2)^2$$

$$49 - 14x + x^2 = 4x^2 - 32x + 64 + 4(y+2)^2$$

$$-15 = 3x^2 - 18x + 4(y+2)^2$$

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$$-15 = 3x^2 - 18x + 4(4+2)^2$$

$$-15 = 3(x^2 - 6x + 9 - 9) + 4(4+2)^2$$

$$-15 = 3(x-3)^2 - 27 + 4(4+2)^2$$

$$12 = 3(x-3)^2 + (4+2)^2$$

$$\frac{(x-3)^2}{4} + \frac{(y+2)^2}{3} = 1$$

Ex (12) Find the foci and equation of the hyperbola with vertices $(0, \pm 1)$ and asymptote $y = 2x$

Formulas (I) Foci $(\pm c, 0)$

Vertices $(\pm a, 0)$

$$c^2 = a^2 + b^2 \quad y = \pm \frac{b}{a}x$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(II) Foci $(0, \pm c)$ $c^2 = a^2 + b^2$ $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

Vertices $(0, \pm a)$

Asymptotes $y = \pm a/b x$

Vertices $(0, \pm 1)$ asymptotes $y = \pm a/b x$ so $b = 1/2$

$$c^2 = 1 + \frac{1}{4} \quad c^2 = \frac{5}{4} \quad c = \pm \frac{\sqrt{5}}{2} \text{ Then, we get the}$$

equation

$$\frac{y^2}{1} - \frac{x^2}{1/4} = 1$$

$$y^2 - 4x^2 = 1$$

Ex 12 What does the following curve represent?

Obtain the equation and sketch the graph

$$9x^2 - 4y^2 - 72x + 8y + 176 = 0$$

$$9(x^2 - 8x) - 4(y^2 - 2y) = -176$$

$$4(y^2 - 2y) - 9(x^2 - 8x) = 176$$

$$4(y^2 - 2y + 1 - 1) - 9(x^2 - 8x + 16 - 16) = 176$$

$$4(y-1)^2 - 4 - 9(x-4)^2 + 144 = 176$$

$$\frac{4(y-1)^2}{9} - \frac{(x-4)^2}{4} = 1$$

Center (4, 1)

$$c^2 = a^2 + b^2 \quad c^2 = 4 + 9 \quad c = \pm\sqrt{13}$$

$$a^2 = 9 \quad a = \pm 3 \quad \text{Major axis } y$$

So, we will add $\pm\sqrt{13}$ to y component.

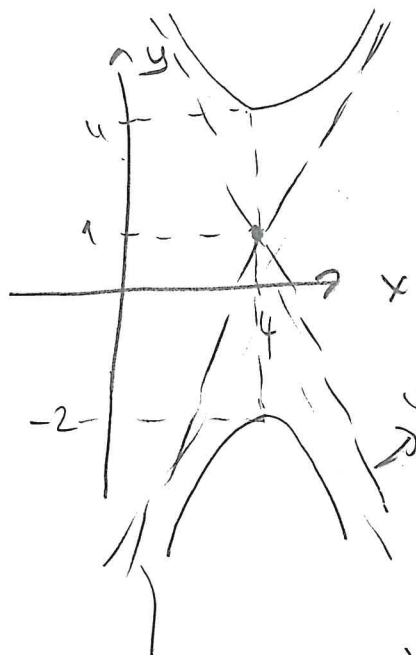
$$\text{Foci } (4, 1 + \sqrt{13}) \quad (4, 1 - \sqrt{13})$$

$$\text{Vertices } (4, 1 + 3) = (4, 4)$$

$$(4, 1 - 3) = (4, -2)$$

Asymptotes

$$y - 1 = \pm \frac{3}{2}(x - 4)$$



$$y - 1 = \frac{3}{2}(x - 4)$$

Ex 13 Identify and sketch the set of points.

Q $x^2 + 2x - y = 3$

Discriminant Test $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

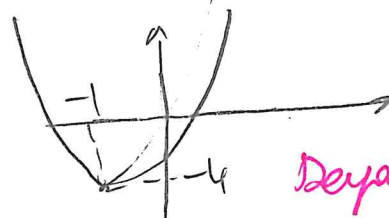
$B^2 - 4AC = 0$ Parabola, $B^2 - 4AC > 0$ Hyperbola, $B^2 - 4AC < 0$

$A = C$ Circle $A \neq C$ Ellipse.

$A = 1 \quad B = 0 \quad C = 0 \quad B^2 - 4AC = 0$ Parabola

$$(x+1)^2 - y = 4$$

$$y = (x+1)^2 - 4$$



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⑥ $9x^2 + 4y^2 - 18x + 8y = -13$

$A=9 \quad B=0 \quad C=4 \quad B^2 - 4AC = 0 - 144 < 0$ Ellipse.

$9(x^2 - 2x + 1 - 1) + 4(y^2 + 2y + 1 - 1) = -13$

$9(x-1)^2 - 9 + 4(y+1)^2 - 4 = -13$

$9(x-1)^2 + 4(y+1)^2 = 0 \Rightarrow$ it is just a single point $(1, -1)$.

⑦ $9x^2 + 4y^2 - 18x + 8y = 23$

$A=9 \quad B=0 \quad C=4 \quad B^2 - 4AC = 0 - 144 < 0$ Ellipse.

$9(x^2 - 2x + 1 - 1) + 4(y^2 + 2y + 1 - 1) = 23$

$9(x-1)^2 - 9 + 4(y+1)^2 - 4 = 23$

$9(x-1)^2 + 4(y+1)^2 = 36$

$\frac{(x-1)^2}{4} + \frac{(y+1)^2}{9} = 1$

$a^2 = 4 \quad a = 2$
 $b^2 = 9 \quad b = 3$

Center $(1, -1)$

$c^2 = a^2 - b^2$

Foci $(1, -1 - \sqrt{5})$

$c^2 = 9 - 4 = 5 \quad c = \pm\sqrt{5}$

$(1, -1 + \sqrt{5})$

Vertices $(1, -1 + 3) = (1, 2)$
 $(1, -1 - 3) = (1, -4)$

Covectors $(1 + 2, -1) = (3, -1)$

$(1 - 2, -1) = (-1, -1)$

⑧ $x^2 - 2y^2 + 3x + 4y = 2$

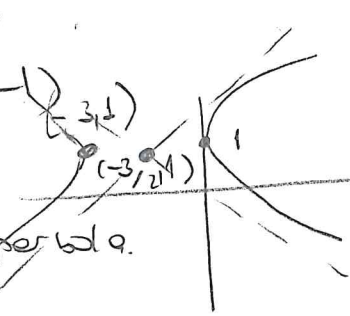
$A=1 \quad B=0 \quad C=-2 \quad B^2 - 4AC = 8 > 0$ Hyperbola.

$x^2 + 3x + \frac{9}{4} - \frac{9}{4} - 2(y^2 - 2y + 1 - 1) = 2$

$(x + \frac{3}{2})^2 - 2(y-1)^2 = \frac{9}{4}$

$\frac{(x + \frac{3}{2})^2}{9/4} - \frac{(y-1)^2}{9/8} = 1$

Center $(-3/2, 1)$ As $2x+3 = \pm 2\sqrt{2}(y-1)$



Ex 14 Find the distance between the points with polar coordinates

$(1, \pi/6)$ and $(3, 3\pi/4)$

$$(r, \theta) = (1, \pi/6) \Rightarrow x = r \cos \theta \quad y = r \sin \theta \quad \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

$$x = 1 \cdot \cos \pi/6 \quad y = 1 \cdot \sin \pi/6 \quad x = \frac{\sqrt{3}}{2} \quad y = \frac{1}{2}$$

$$P_1 = (x_1, y_1) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

$$(r, \theta) = (3, 3\pi/4) \Rightarrow x = 3 \cdot \cos 3\pi/4 = -\frac{3\sqrt{2}}{2} \quad P_2 = (x_2, y_2) = \left(-\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$$

$$y = 3 \sin 3\pi/4 = \frac{3\sqrt{2}}{2}$$

Distance between P_1 and P_2

$$\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = \sqrt{\left(\frac{\sqrt{3}}{2} + \frac{3\sqrt{2}}{2}\right)^2 + \left(\frac{1}{2} - \frac{3\sqrt{2}}{2}\right)^2} = \frac{1}{2} \sqrt{10 + 6\sqrt{6} - 6\sqrt{2}}$$

Ex 15 Find the formula for the distance between points

(r_1, θ_1) (r_2, θ_2)

$$x_1 = r_1 \cos \theta_1 \quad y_1 = r_1 \sin \theta_1 \quad P_1 = (r_1 \cos \theta_1, r_1 \sin \theta_1)$$

$$x_2 = r_2 \cos \theta_2 \quad y_2 = r_2 \sin \theta_2 \quad P_2 = (r_2 \cos \theta_2, r_2 \sin \theta_2)$$

$$|P_1 P_2| = \sqrt{(r_1 \cos \theta_1 - r_2 \cos \theta_2)^2 + (r_1 \sin \theta_1 - r_2 \sin \theta_2)^2}$$

$$= \sqrt{r_1^2 \cos^2 \theta_1 - 2r_1 r_2 \cos \theta_1 \cos \theta_2 + r_2^2 \cos^2 \theta_2}$$

$$+ r_1^2 \sin^2 \theta_1 - 2r_1 r_2 \sin \theta_1 \sin \theta_2 + r_2^2 \sin^2 \theta_2$$

$$= \sqrt{r_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) - 2r_1 r_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + r_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2)}$$

$$= \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)}$$

Ex 16 Identify the following curves using Cartesian coordinates

a) $r = 2 \sin \theta + 2 \cos \theta$

Multiply both sides with r

$$x^2 + y^2 = 2y + 2x$$

$$r^2 = \frac{2r \sin \theta}{y} + \frac{2r \cos \theta}{x}$$

$$x^2 - 2x + 1 - 1 + y^2 - 2y + 1 - 1 = 0$$

$$(x-1)^2 + (y-1)^2 = 2$$

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M(1,1) $r = \sqrt{2}$
center.

(13)

④ $r = \tan \theta \sec \theta$

$$r = \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta} = \frac{\sin \theta}{\cos^2 \theta}$$

Consider $\cos \theta \neq 0$

$$r \cdot \cos^2 \theta = \sin \theta \Rightarrow$$

Multiply both sides with r .

$$\underbrace{r^2 \cos^2 \theta}_{x^2} = \underbrace{r \sin \theta}_y \quad x^2 = y$$

$\cos \theta = 0$ gives $r \cos^2 \theta = \sin \theta = 0$ which is a contradiction with

$$\cos^2 \theta + \sin^2 \theta = 1.$$

So, we will only consider the case $\cos \theta \neq 0$.

Ex 17 Find a polar equation for the following curves

① $x = -y^2$

② $x + y = 9$

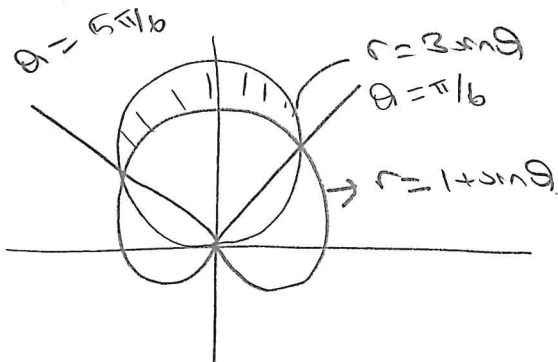
① $x = -y^2 \Rightarrow r \cos \theta = -r^2 \sin^2 \theta \Rightarrow \cos \theta = -r \sin^2 \theta$

$$r = -\frac{\cos \theta}{\sin^2 \theta} = -\cot \theta \sec \theta$$

② $x + y = 9 \Rightarrow r \cos \theta + r \sin \theta = 9 \Rightarrow r = 9 / (\cos \theta + \sin \theta)$

Ex 18 Find the area of the region that lies inside the circle

$r = 3 \sin \theta$ and outside of the cardioid $r = 1 + \sin \theta$



$$3 \sin \theta = 1 + \sin \theta \quad \sin \theta = 1/2 \quad \theta = \pi/6, 5\pi/6$$

$$A = \frac{1}{2} \int_{\pi/6}^{5\pi/6} (3 \sin \theta)^2 d\theta - \frac{1}{2} \int_{\pi/6}^{5\pi/6} (1 + \sin \theta)^2 d\theta$$

The area of the circle

Use symmetry

between $\pi/6 \leq \theta \leq 5\pi/6$

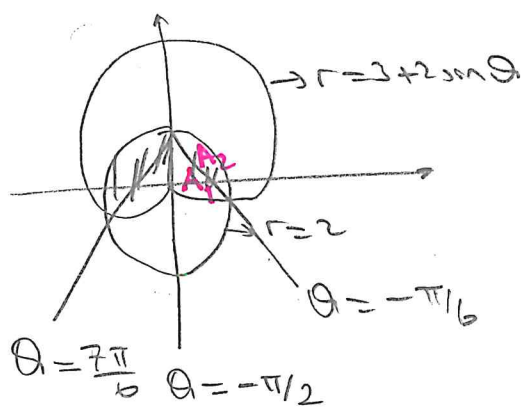
$$A = 2 \left[\frac{1}{2} \int_{\pi/6}^{\pi/2} 9 \sin^2 \theta d\theta - \frac{1}{2} \int_{\pi/6}^{\pi/2} (1 + 2 \sin \theta + \sin^2 \theta) d\theta \right]$$

$$= \int_{\pi/6}^{\pi/2} (8 \sin^2 \theta - 1 - 2 \sin \theta) d\theta = \int_{\pi/6}^{\pi/2} (3 - 4 \cos 2\theta - 2 \sin \theta) d\theta$$

Pay Attention

$$\textcircled{14} \quad \left. 3\theta - 2 \sin 2\theta + 2 \cos \theta \right|_{\pi/6}^{\pi/2} = \pi$$

Ex 17 Find the area of the region that lies inside both curves



$$3 + 2 \sin \theta = 2$$

$$2 \sin \theta = -1$$

$$\sin \theta = -1/2$$

$$\theta = -\pi/6$$

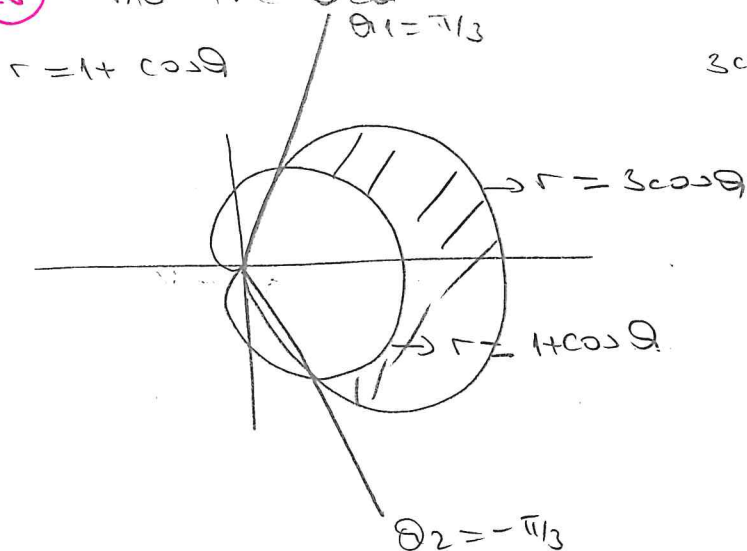
The graph is symmetric so $+\pi/2$

$$A = 2 \left[\int_{-\pi/2}^{-\pi/6} \frac{1}{2} (3 + 2 \sin \theta)^2 d\theta + \int_{-\pi/6}^{+\pi/2} \frac{1}{2} 2^2 d\theta \right]$$

$$A = 2 \left[\int_{-\pi/2}^{-\pi/6} (9 + 12 \sin \theta + 4 \sin^2 \theta) d\theta + 4\theta \Big|_{-\pi/6}^{+\pi/2} \right] \quad \text{A}_1 \text{ inside cardioid.} \quad \text{A}_2 \text{ inside circle.}$$

$$= \left[9\theta - 12 \cos \theta + 2\theta - \sin 2\theta \right]_{-\pi/2}^{-\pi/6} + \frac{8\pi}{3} = \frac{19\pi}{3} - \frac{11\sqrt{3}}{2}$$

Ex 18 Find the area that lies inside $r = 3 \cos \theta$ and outside



$$3 \cos \theta = 1 + \cos \theta$$

$$2 \cos \theta = 1$$

$$\cos \theta = 1/2 \quad \theta_1 = \pi/3$$

$$\theta_2 = -\pi/3$$

Use symmetry

$$A = 2 \int_0^{\pi/3} \frac{1}{2} [(3 \cos \theta)^2 - (1 + \cos \theta)^2] d\theta$$

$$= \int_0^{\pi/3} [8 \cos^2 \theta - 2 \cos \theta - 1] d\theta$$

$$= \int_0^{\pi/3} (4(1 + \cos 2\theta) - 2 \cos \theta - 1) d\theta$$

$$= [3\theta + 2 \sin 2\theta - 2 \sin \theta] \Big|_0^{\pi/3} = \pi + \sqrt{3} - \sqrt{3} = \pi$$

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Ex. 19 Find the exact length of $r = 3 \sin \theta$ $0 \leq \theta \leq \pi/3$

For $r = f(\theta)$ $a \leq \theta \leq b$

$$L = \int_a^b \sqrt{r^2 + (r')^2} d\theta$$

$$r = 3 \sin \theta \quad r' = 3 \cos \theta$$

So, we have $L = \int_0^{\pi/3} \sqrt{(3 \sin \theta)^2 + (3 \cos \theta)^2} d\theta =$

$$L = \int_0^{\pi/3} \sqrt{9(\sin^2 \theta + \cos^2 \theta)} d\theta = 3 \cdot \pi/3 = \pi.$$

Ex. 20 Sketch the curve $r = 2 \cos 3\theta$ and find the area that it encloses

$$r = 2 \cos 3\theta$$

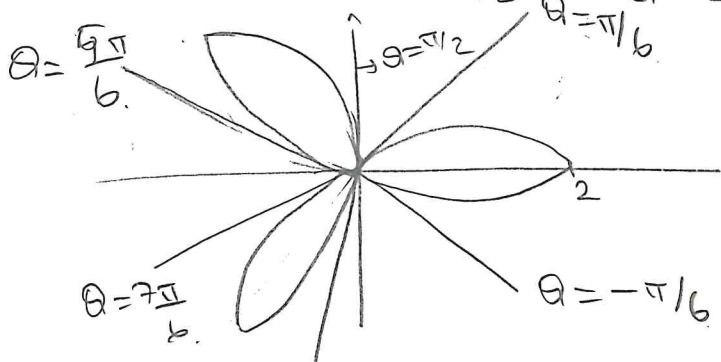
$$\cos 3\theta = 0 \quad \theta = 3\theta = \pi/2$$

$$3\theta = 7\pi/2$$

$$\theta = \pi/6$$

$$\theta = -\pi/6$$

r	θ
2	0
0	$\pi/6$



$$\text{Area} = 6 \int_0^{\pi/6} \frac{1}{2} (2 \cos 3\theta)^2 d\theta \quad \theta = \frac{11\pi}{6}$$

$$= 12 \int_0^{\pi/6} \cos^2 3\theta d\theta$$

$$= \frac{12}{2} \int_0^{\pi/6} (1 + \cos 6\theta) d\theta$$

$$= 6 \left[\theta + \frac{1}{6} \sin 6\theta \right]_0^{\pi/6} = \pi$$

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